AN AFFINE MORPHISM IS LOCAL COMPLETE INTERSECTION EXACTLY WHEN ITS RING MAP IS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18177-18184 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18186-18188, the complete author proof (1 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE for Chief: proof is 'follows from the definitions' with no cross-chapter citation in the proof body (the only reference sits inside the statement). Kept under the same reading as wave 16's accepted dictionary cards; add to the pending BORDERLINE tally (would make 9).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Schemes lci morphism definition (tags chapter, in-source); More on Algebra, definition local-complete-intersection (cited by label in the statement).

Source reference. Original lemma 07D0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18188–18188. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X = \mathrm{Spec}(B) \to S = \mathrm{Spec}(A)$ be a morphism of affine schemes. Then f is a local complete intersection morphism if and only if $A \to B$ is a local complete intersection homomorphism, see More on Algebra, Definition 07D0.*

Proof. Follows immediately from the definitions. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.